MODULE 07 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

AI Architecture: RAG and Inference

Grounding answers in your own documents, and hosting models that Bedrock does not serve.

DOMAIN 1 of 5 EXAM WEIGHT 25% MODULE 7 of 52

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RAG does not change the model. It changes **what the model can see** at the moment it answers.

RAG or a customised model?

CHOOSE RAG WHEN

The problem is knowledge

- Facts the model was never trained on
- Content that changes often
- Answers must cite their sources
- Access depends on who is asking

CONSIDER CUSTOMISATION WHEN

The problem is behaviour

- A specific tone or output format
- A narrow task done consistently
- Labelled examples are available
- Model training is out of exam scope

THE TIEBREAKER

Changing facts and citations mean RAG. Customising a model is the wrong tool for knowledge that changes weekly.

The RAG pipeline

01
Connect
S3, SharePoint, Confluence and others, synced.

02
Parse
Default parser, a foundation model, or Data Automation.

03
Chunk
Split documents into retrievable passages.

04
Embed
An embeddings model turns each chunk into a vector.

05
Store
A vector index holds vectors and metadata.

06
Retrieve
Find, filter, rerank, then generate with citations.

ON THE EXAM

Skill 1.2.4 says ingestion and retrieval strategies. Most questions are about chunking, storage or filtering.

Managed or custom knowledge base?

MANAGED KNOWLEDGE BASE

AWS runs the pipeline

- GA since June 2026
- Ingestion, vectors, storage and retrieval managed
- Six connectors, including SharePoint and S3
- Agentic retrieval for multi-step questions

CUSTOM KNOWLEDGE BASE

You choose each piece

- Pick the vector store
- Choose parsing and chunking strategy
- GraphRAG with Neptune Analytics
- Reuse an existing vector database

THE TIEBREAKER

Least operational overhead, no vector-store requirement: Managed. A named vector store or chunking need: custom.

Vector stores for a custom knowledge base

STORE	CHOOSE IT WHEN
OpenSearch Serverless	Default; hybrid search; needed for SharePoint, Confluence, Salesforce sources
Aurora PostgreSQL	Vectors beside relational data you already run
Neptune Analytics	GraphRAG: answers depend on relationships
Amazon S3 Vectors	Very large, cost-sensitive, sub-second is fast enough
Third-party stores	An existing Pinecone, MongoDB Atlas or Redis estate

ON THE EXAM

S3 Vectors, GA December 2025, claims up to 90 percent lower cost. The trade is latency, not durability.

S3 Vectors or OpenSearch Serverless?

CHOOSE S3 VECTORS WHEN

Scale and cost first

- Billions of vectors
- Queries are occasional
- Under a second is acceptable
- Lowest cost to store and query

CHOOSE OPENSEARCH WHEN

Latency and features first

- High query rates
- Consistent low latency
- Hybrid keyword and vector search
- Rich filtering and aggregations

THE TIEBREAKER

Large and rarely queried means S3 Vectors. Busy, interactive and hybrid means OpenSearch.

Chunking strategies

STRATEGY	WHAT IT DOES	USE IT WHEN
Default	About 300 tokens, keeps sentences	No special structure
Fixed-size	Set token count and overlap	Uniform documents
Hierarchical	Small child chunks inside larger parents	Precise match, wide context
Semantic	Splits where meaning changes	Topics shift within documents
None	Each document is one chunk	Already split, such as FAQs
Custom Lambda	Your logic after chunking	Special formats or metadata

ON THE EXAM

The chunking strategy cannot be changed after the data source is connected. Choose before the first sync.

Levers for retrieval quality

- **Metadata filtering** narrows search by tenant, department, date or document type.
- **Hybrid search** adds keyword matching for part numbers, codes and names.
- **Reranking models** reorder results so the best passages reach the model.
- **Agentic retrieval** splits multi-part questions into sub-queries.
- **Structured data:** GenerateQuery turns questions into SQL for Amazon Redshift.

> ON THE EXAM
>
> Exact identifiers that semantic search misses, such as order numbers, are a hybrid search signal.

Retrieve or RetrieveAndGenerate?

RETRIEVE

You orchestrate

- Returns the relevant chunks only
- You build the prompt
- Any model, any framework
- Fits agents that decide what to do

RETRIEVEANDGENERATE

Bedrock orchestrates

- Retrieval and generation in one call
- Citations to source chunks
- Least code
- Guardrails can be applied

THE TIEBREAKER

Least code with citations: RetrieveAndGenerate. Custom prompting or an agent framework: Retrieve.

Securing a RAG system

- **Retrieval returns whatever was ingested.** It does not know who is asking.
- **Apply the filter server-side** from the caller's verified identity, never from user input.
- **Separate knowledge bases** when isolation must be physical, as in a silo tier.
- **Customer managed KMS keys** for the vector store and multimodal storage.
- **Guardrails** on generation, including grounding checks against retrieved text.

> ON THE EXAM
>
> A prompt telling the model to ignore other tenants' documents is not access control.

SageMaker AI inference options

OPTION	LIMITS	CHOOSE IT WHEN
Real-time endpoint	25 MB, 60 s; 8 min streaming	Steady, low-latency traffic
Serverless inference	4 MB, 60 s; pay per use	Intermittent, unpredictable traffic
Asynchronous inference	1 GB, up to 1 hour; scales to zero	Large inputs, long processing, queued
Batch transform	Datasets of GBs; runs for days	Offline scoring, no endpoint needed

ON THE EXAM

Payload size and processing time decide this table. A 300 MB input or a 20-minute job means asynchronous.

Bedrock or a SageMaker AI endpoint?

CHOOSE BEDROCK WHEN

A foundation model by API

- Model is offered on Bedrock
- No infrastructure to manage
- Pay per token or by tier
- Guardrails, knowledge bases, agents built in

CHOOSE SAGEMAKER AI WHEN

Your model, your hosting

- A model you trained or an open-weight model
- Custom container or framework
- Choose instance and GPU types
- Classic ML: forecasting, classification

THE TIEBREAKER

A model the company built itself, or one Bedrock does not offer, needs a SageMaker AI endpoint.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Content changes weekly; cite sources	RAG, not a customised model
RAG with least operational overhead	Managed Knowledge Base
Billions of vectors, rarely queried, cheapest	Amazon S3 Vectors
Accurate match, but answers lack context	Hierarchical chunking
User saw another tenant's document	Server-side metadata filter
Own model, huge inputs, long jobs	SageMaker asynchronous inference

ON THE EXAM

RAG questions are rarely about the model. They are about data: how it is cut, stored, filtered and secured.

> Meridian Freight wants an assistant that answers customer questions about its shipping policies. Policies change most weeks, and legal requires every answer to cite the policy section it came from.
>
> The company has no machine learning team.

What should the solutions architect recommend?

A. Fine-tune a foundation model on the policy documents every week

B. A Bedrock knowledge base over the policies, queried with RetrieveAndGenerate

C. Include every policy document in the system prompt of each request

D. Run continued pre-training on the policies each quarter

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Weekly fine-tuning

A training run per policy change, needs ML skills, and cannot cite sources.

B Knowledge base with RetrieveAndGenerate

Policies re-sync as they change, and answers cite the chunks used.

C Everything in the prompt

Pays for every policy on every request, and hits context limits as policies grow.

D Quarterly pre-training

Knowledge would be months out of date, and still no citations.

THE SIGNAL

Weekly change plus mandatory citations. Both point at retrieval, and neither can be met by training.

> Meridian Freight's operations team wants a RAG assistant over procedures held in SharePoint and in an S3 bucket. The team has no experience running vector databases.
>
> It has no requirement for a particular vector store or chunking approach, and wants the LEAST operational overhead.

Which solution should the solutions architect recommend?

A. A custom knowledge base with a quick-created OpenSearch Serverless collection

B. A self-managed vector database on EC2, populated by a Lambda pipeline

C. An Amazon Bedrock Managed Knowledge Base with SharePoint and S3 connectors

D. An Amazon Kendra index connected to SharePoint

◆ PAUSE AND DECIDE

Which solution should the solutions architect recommend?

A Custom KB, OpenSearch Serverless

Works, and was the answer before June 2026. The team still owns a vector store.

B Self-managed vector database

The most to build and run, for a team with no experience.

C Managed Knowledge Base

AWS runs ingestion, vectors, storage and retrieval; both sources have native connectors.

D Amazon Kendra

Not on this exam's in-scope service list.

THE SIGNAL

No vector store requirement and least overhead. Since June 2026 that describes a Managed Knowledge Base.

> Meridian Freight is building semantic search over 15 years of archived shipping documents: about 800 million embeddings. Auditors run a few hundred searches a day, and a response within a second is acceptable.
>
> The company uses a custom knowledge base and wants the lowest storage and query cost.

Which vector store should the solutions architect choose?

A Amazon OpenSearch Serverless

B Amazon Aurora PostgreSQL with pgvector

C Amazon Neptune Analytics

D Amazon S3 Vectors

◆ PAUSE AND DECIDE

Which vector store should the solutions architect choose?

A OpenSearch Serverless

Fast and capable, but costly to keep running for a few hundred queries a day.

B Aurora with pgvector

A provisioned database for an archive that is mostly idle.

C Neptune Analytics

For relationship-heavy GraphRAG. Nothing here needs a graph.

D Amazon S3 Vectors

Built for very large, rarely queried indexes at the lowest cost.

THE SIGNAL

Huge, rarely queried, sub-second acceptable, cheapest. Every clause describes S3 Vectors.

> Meridian Freight's knowledge base over long equipment manuals uses small fixed-size chunks. Retrieval finds the right passage, but answers often omit warnings and steps from the surrounding section.
>
> The team wants precise matching with more context in each answer.

What should the solutions architect change?

A Re-ingest with hierarchical chunking, searching child chunks and returning their parents

B Switch to no chunking, so each manual is one chunk

C Increase the number of results returned from 5 to 50

D Add a guardrail with a contextual grounding check

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Hierarchical chunking

Small children keep the match precise; the parent supplies the section.

B No chunking

A whole manual per chunk destroys match precision and floods the context.

C Fifty results

Adds many unrelated passages rather than the neighbouring ones.

D Grounding check

Flags unsupported answers. It cannot add context that was never retrieved.

THE SIGNAL

Retrieval is accurate but context is missing. That exact pair of symptoms is what hierarchical chunking solves.

> Meridian Freight's SaaS platform uses one knowledge base for all customers' uploaded documents. Each chunk was ingested with a tenant_id metadata attribute, but retrieval does not use it.
>
> A user from one customer received a passage from another customer's contract.

What should the solutions architect do?

A. Add an instruction to the prompt to use only the user's own company's documents

B. Apply a tenant_id metadata filter, taken from the verified token, on every retrieval request

C. Add a guardrail denied topic for other customers' names

D. Add a reranking model to improve relevance

◆ PAUSE AND DECIDE

What should the solutions architect do?

A **Prompt instruction**

The passage has already been retrieved. A prompt is not access control.

B **Server-side metadata filter**

Other tenants' chunks are never retrieved, so they cannot leak.

C **Guardrail denied topic**

Matches topics, not ownership; leaks anything not on the list.

D **Reranking**

Improves ordering. Another tenant's passage can still be the best match.

THE SIGNAL

The metadata already exists. The fix is to filter on it, from identity the user cannot alter.

> Meridian Freight's data science team has trained a damage-assessment model. Each input is a bundle of photographs of about 300 MB, and each assessment takes around 20 minutes.
>
> Claims arrive unpredictably, sometimes none for days. The company wants to pay as little as possible while idle.

How should the model be hosted?

A. A SageMaker AI real-time endpoint

B. SageMaker AI serverless inference

C. SageMaker AI asynchronous inference, scaled to zero when idle

D. An on-demand Amazon Bedrock model

◆ PAUSE AND DECIDE

How should the model be hosted?

A **Real-time endpoint**

25 MB and 60-second limits, and instances run while idle.

B **Serverless inference**

4 MB and 60 seconds. Both far below this workload.

C **Asynchronous, scale to zero**

1 GB inputs, up to an hour each, and no instances while idle.

D **Bedrock on-demand**

Bedrock serves its catalogue of models. This is the company's own model.

THE SIGNAL

300 MB and 20 minutes exceed every option except asynchronous. Idle days make scale to zero the clincher.

What to carry out of this module

- **Knowledge problem, RAG.** Behaviour problem, customisation, which is out of exam scope.
- **Managed Knowledge Base** since June 2026 for least overhead; custom for specific stores.
- **S3 Vectors** for huge, cheap, rarely queried; **OpenSearch** for busy and hybrid.
- **Hierarchical chunking** when matches are right but context is thin.
- **Filter retrieval by verified identity.** Prompts are not access control.

> ON THE EXAM
>
> For SageMaker AI, check payload and duration against each option's limits before anything else.